

TWINKLE, TWINKLE LITTLE LEDs

The LED industry has much reason to thank the material sciences and semiconductor fabrication worlds, which have been responsible for many of the significant improvements in this space, especially related to efficacy and cost. As a result, LEDs have found lots more uses, and are slated to takeover the large domestic market as well



OSRAM brilliant mix concept

■ JANANI GOPALAKRISHNAN VIKRAM

Big things come in small packages. This statement holds 100 per cent true for light-emitting diodes (LEDs), which are now celebrated for many salient features including small form factor, long life and low power consumption.

That said, you might be surprised to learn that the first LEDs were far from being useful. They could only operate at bizarre temperatures of -77°C and below, would burn out in only a few hours, guzzle power like demons, and produce very little light in the bargain. As we can see, all of these cited weaknesses have over time transformed into strengths—mainly due to process and material improvements. In short, the technological and business developments in the LED industry have been, and continue to be, evolutionary and revolutionary at the same time.

In the past few years, improvements in efficacy, packaging and price of LEDs have resulted in a greater realisation of its benefits, more applications and a larger market. "The range of applications for LEDs continues to grow significantly. One of the fastest-growing and most promising uses is in illumination applications. LEDs are penetrating this market rapidly due to the combination of excellent colour